Payton Bontje

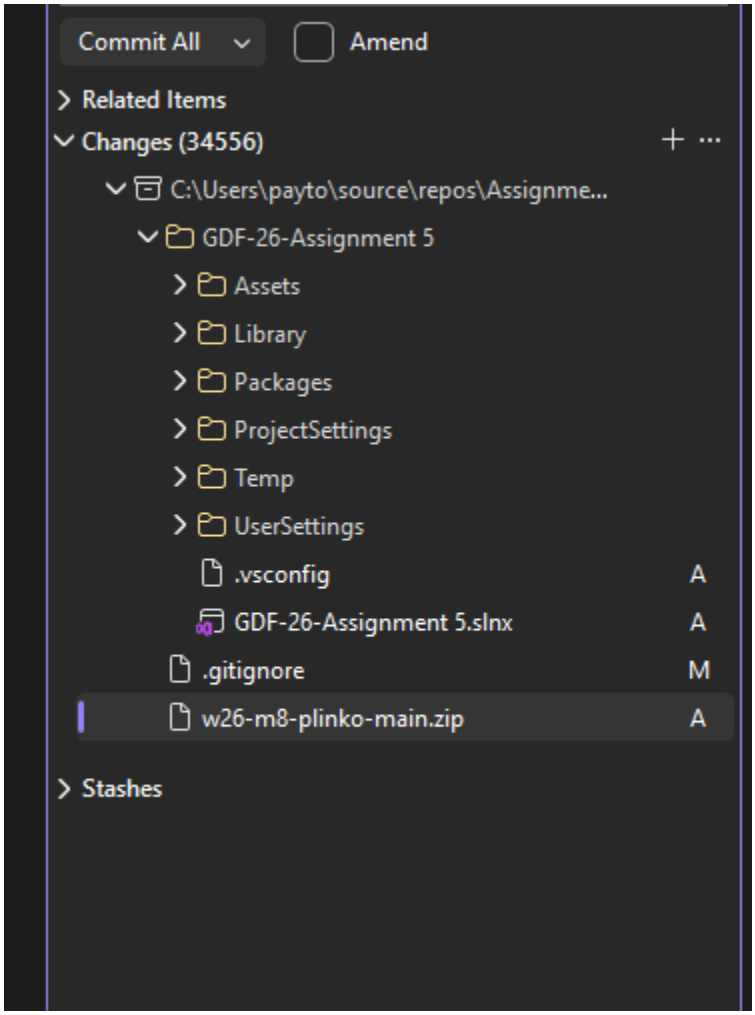
Assignment 5 - Process Work

This time, I'm starting work early.
I don't have a clear idea in mind from the start, and I am well behind everyone else due to Assignment 4 occupying my time in class. I only just downloaded unity tonight (a few days before the assignment is due), and already I am having issues. For some reason, the project files that were on my laptop version of Unity used for an assignment in the 2D assets class have not carried over to my desktop download of Unity.
I tried using multiple emails to check, so it wasn't an email issue. This isn't technically mandatory for getting the work done, but frankly I would prefer to have the ability to work on multiple assignments from the same computer rather than… anyways. Plinko. Pachinko. A slot machine that isn't a slot machine. I'll start there, after reviewing some of the videos.

First issue, downloaded the wrong version. Fixed by simply downloading the right one.
Second issue, figuring out what the second issue even is.

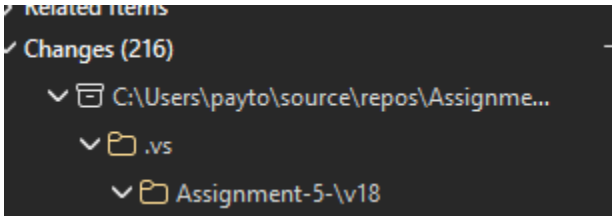
Second issue, watching all the workalong videos I missed.

Second real issue, I'm getting stuck on the 30,000 changes issue.



I think its caused by the fact that unity generated all of the stuff in GDF-26-Assignment 5 instead of using the stuff from the plinko zip folder. Not really sure how to quickly fix that.
The two things I could potentially do are either delete the unzipped plinko stuff, but I think the issue is earlier?

Wait, I think maybe I thought the gitignore being in the plinko zip file was where it needed to be, but when you specifically mentioned putting the gitignore in the file its just because you named it that, when I need to put the unity gitignore in the assignment 5 file.



Yeah that did it.

Ok, 216 is still well beyond what its supposed to be considering the recording had it at 64. Time to get back to that.

Realized that the reason it's much bigger is because I have the plinko zip folder which likely added the extra 150 ish changes.
Currently have managed to open up VS via the scripts thing.

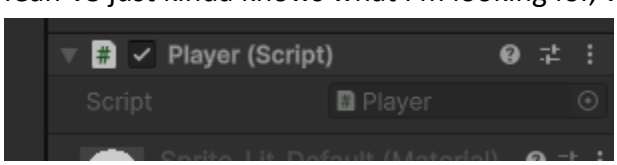
```
// Update is called once per frame
Unity Message | 0 references
void Update()
{

    float movement = Input.GetAxis("Horizontal");
    transform.position = new Vector3(transform.position.x + movement, transform.position.y, movement);

    if (move
    {

    }
```

Very much just going to be copying things until later. Though I'm very much feeling the syntax issue creeping in again. It's the biggest issue I'm having, the 'translation' of code to english. My brain just doesn't work with code as a language yet, so the English pattern recognition takes over seeing words I know but not in a form I know.

```
float movementX = Input.GetAxis("Horizontal");
Vector3 position = transform.position;
position.x = movementX;
position
```

```
float movementX = Input.GetAxis("Horizontal");
Vector3 position = transform.position;
position.x = movementX;
transform.position = position;
```

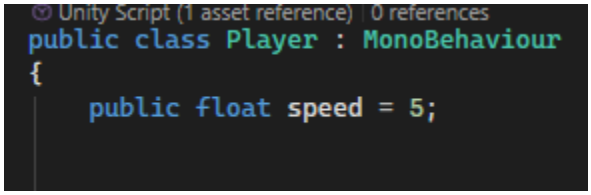
Yeah VS just kinda knows what I'm looking for, which is making this go faster.

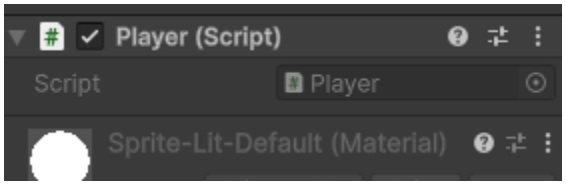


It's not working.

```
InvalidOperationException: You are trying to read Input using the UnityEngine.Input class, but you have switched active Input handling to Input System package in Player Settings.
```

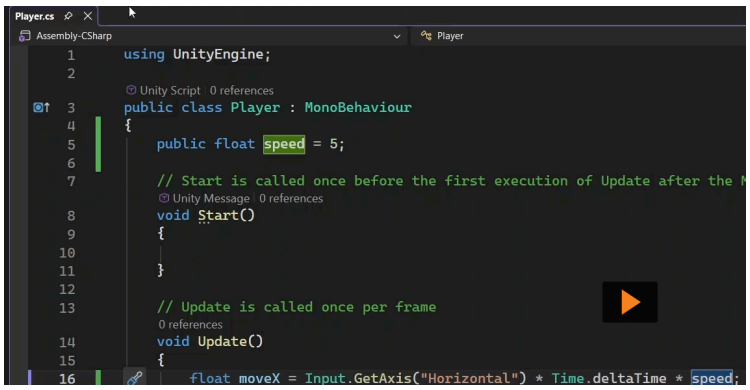
Fuck fuck fuck.
Just gonna google for help. (Google was unhelpful)
Ok, I think I vaguely understand that the script is just not working for zero reason. Ignore the red text that gives a magical syntactical bullshit excuse for its conscious choice to disobey.

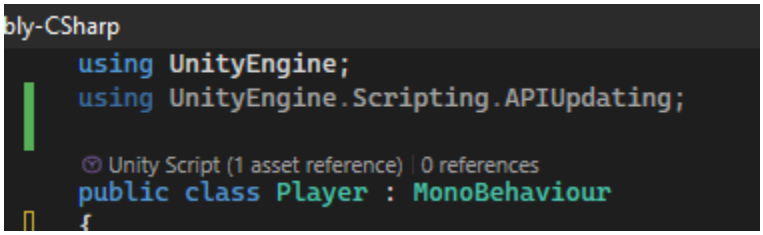




The speed is not showing up, meaning that the script is in some way just not the right one?
But it is the right one, double clicking it just brings me back to the VS code.

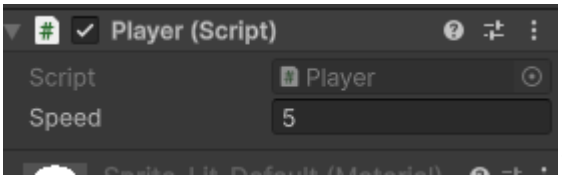
Oh my fucking god I see it. This is the same shit that Anaha helped with for our 2D Assets class.



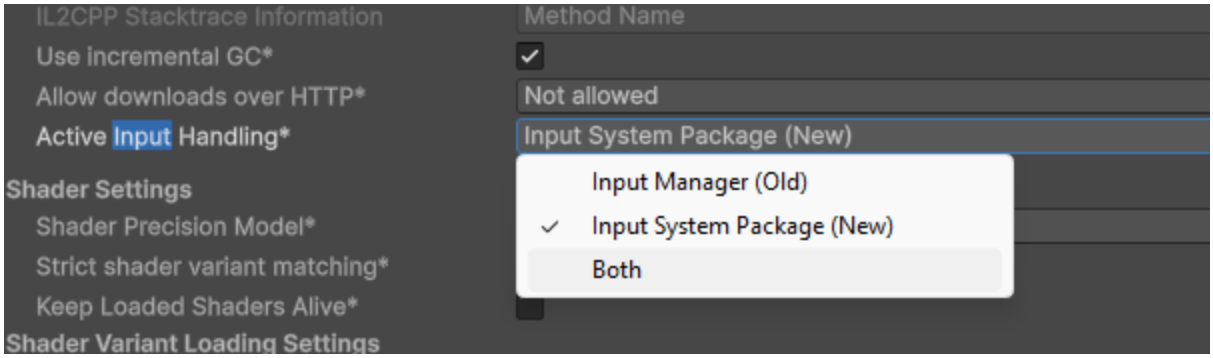


Despite doing nothing differently, it decided to add a second library that conflicts with it, for no reason. Deleting it should fix the problem.
(It didn't).

Ok, no fucking idea what to do again.

????? 

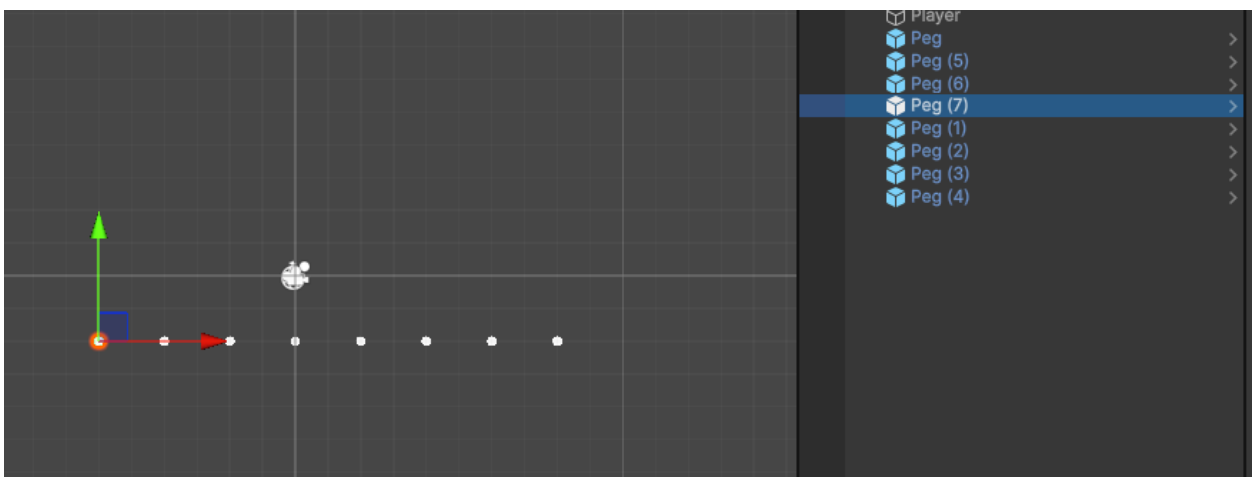
What the actual fuck? It just suddenly showed up correctly? Even more of a ???? it's STILL NOT WORKING.
Fuck it, I'm going to try the other method of checking key presses.

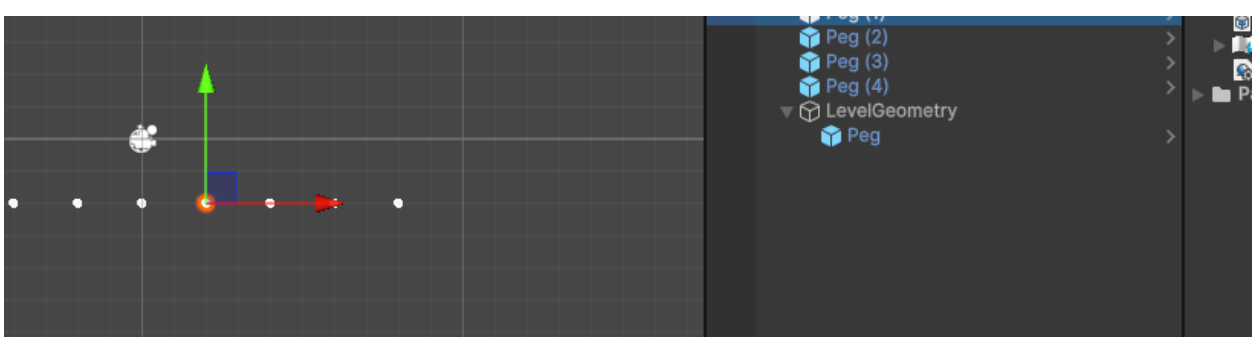


Oh my fucking god. Ok, so this shit has literally taken me like a full 2 hours just to track down the OTHER thing that Anaha showed me was fucking stupid about Unity. Well now that that's fixed… MAYBE?!?!?!?!?

Yeah that fixed it. This would have gone SOOOOO much faster if I had just asked Anaha, because this is literally the issue he already showed me. This is what I get for being stubborn/being unable to work unless it's 1:30 in the morning.

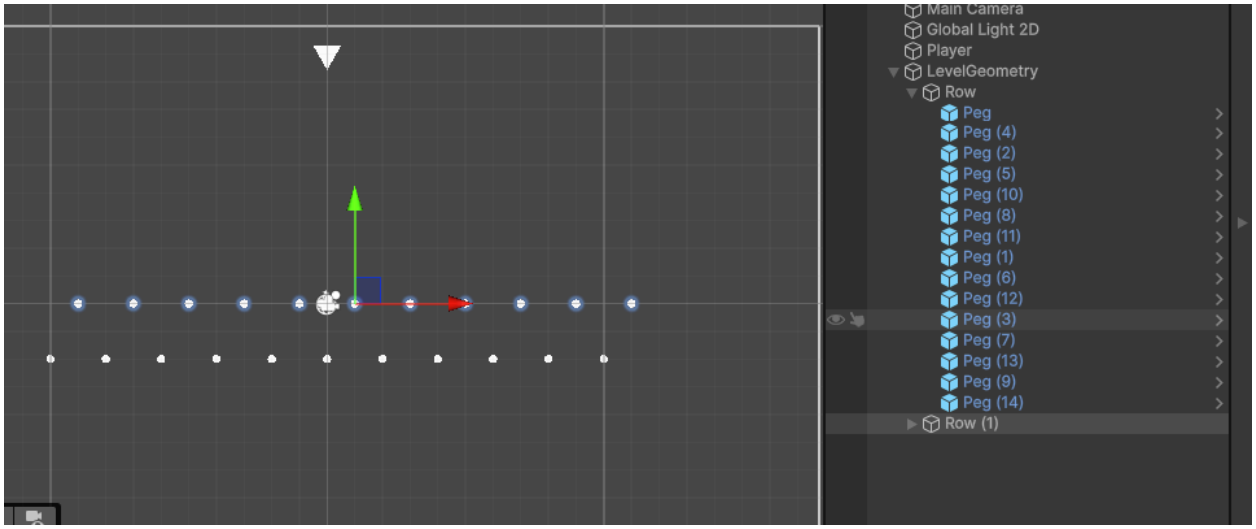
Picking back up, I've begun by pasting a bunch of additional pegs. Gonna put them in a folder or find some other way of grouping them/converting them into a single game object.

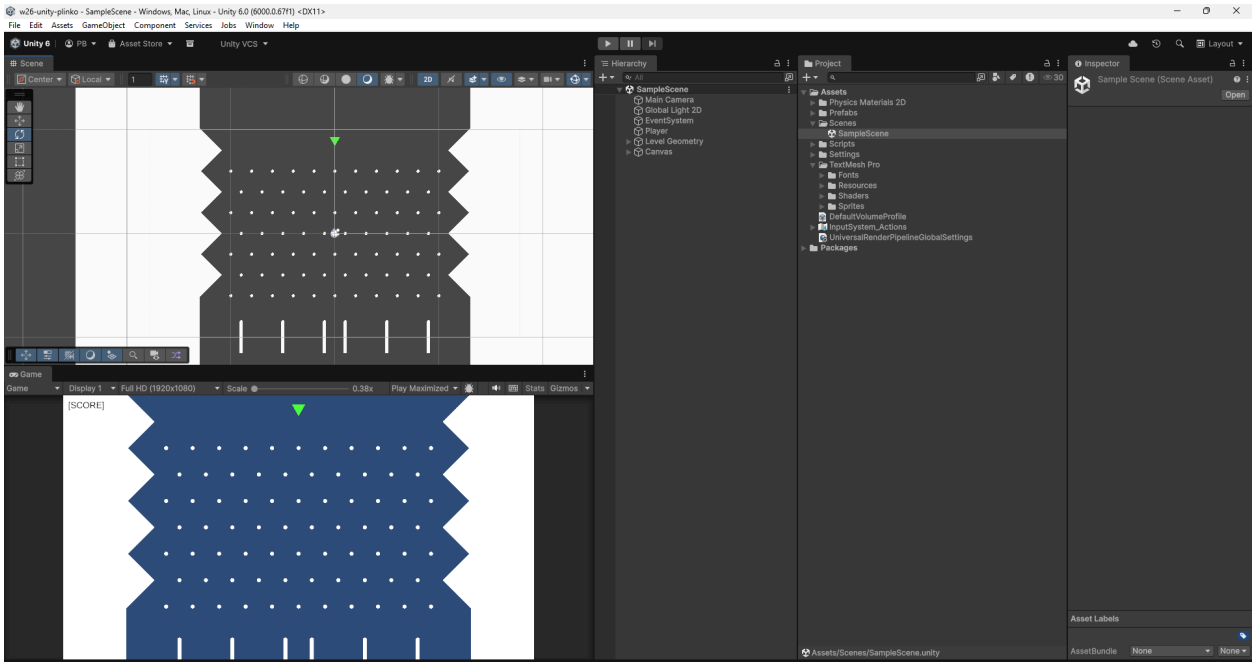




I was getting ahead of myself

Yeah this is exactly what I was looking for.





Ok, Few days have passed, I missed documenting the last few updates to the old version, but that doesn't matter because this shit needs to get done, so I have asked Anaha to help me find the way to open the zipped project folder to catch up quickly.

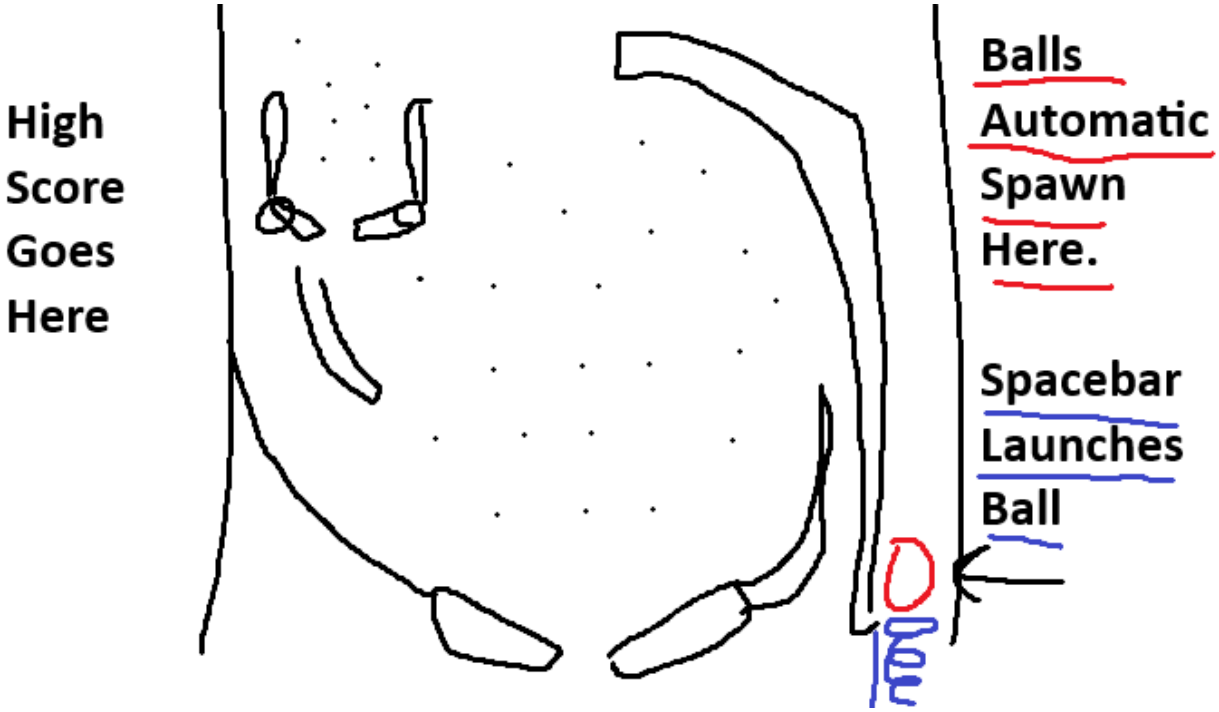
Now is a matter of reading the rubric to figure out what I can most easily mod in to meet the criteria of the assignment. Ambition will have to perish in the face of crushing time demands.

Actually, before I read the rubric, I have to figure out a general thing I want to do. While this doesn't feel spiritually like a Mod, I feel like Pinball is the obvious answer for something that is relatively easy to implement yet covers most of what's expected.

Current Implementation Checklist for things I need:

- Bumpers controlled via arrow left and arrow right (functionality should be easy, movement maybe a little difficult)
- Points for hitting the pins
- A highscore system that tracks which ball had the highest points scored.
- Special zones that if you hit the ball into, give additional points.
- Sound effects
- Music

Realistically, I have like two days left to work on this, and I doubt I can get it all done. The list is sort of organized by importance, and luckily it feels like most of the systems won't take toooo much adapting. Sketch on the next page.



The basic idea is pretty simple, I'm drawing a lot from that old pinball game that used to be automatically included in old PC's/Laptops.

Can you tell I made my diagram before grabbing the reference image? Though I swear I remember there being a version that featured a miniature version of pinball off to the side.

Initial work on the ball launcher – I do not know how to write the kind of movement code I am trying to write. What I need to do is have some way to make the script move the launcher a pre-defined amount of distance, then have it reset when the spacebar isn't actively being pressed.

```
Player.cs    Ball Launcher.cs*
Assembly-CSharp    BallLauncher    Start()
using UnityEngine;

Unity Script | 0 references
public class BallLauncher : MonoBehaviour
{
    public float LaunchSpeed = 20.0f;
    float moveY;
    Vector3 position;
    private Vector3 startPosition;
    private Quaternion startRotation;

    // Start is called once before the first execution of Update after the MonoBehaviour is created
    Unity Message | 0 references
    void Start()
    {

    }



    // Update is called once per frame
    Unity Message | 0 references
    void Update()
    {
        // This is copied and changed from the initial player script, to (hopefully) allow for
        // launching the ball via a movement and collision script with the ball.
        // Launch Player




        // This should allow the ball to spawn when spacebar is pressed, then have the ball launch
        // when spacebar is released.
        if (Input.GetButtonUp(KeyCode.Space))
        {
            moveY = position.y *Time.deltaTime * LaunchSpeed;
            transform.position =
            position.x += moveY;
            transform.position = position;
```

It should be the same as doing the movement calculation during the assignment where I made the balls move… I think???? I know its a different framework but surely the principal is the same.

:(

```
// This should allow the ball to spawn when spacebar is pressed, then have the ball launch
// when spacebar is released.
if (Input.GetButtonUp(KeyCode.Space))
{
    moveY = position.y *Time.de
    transform.position =

    transform.position = positi

}
else
{
```

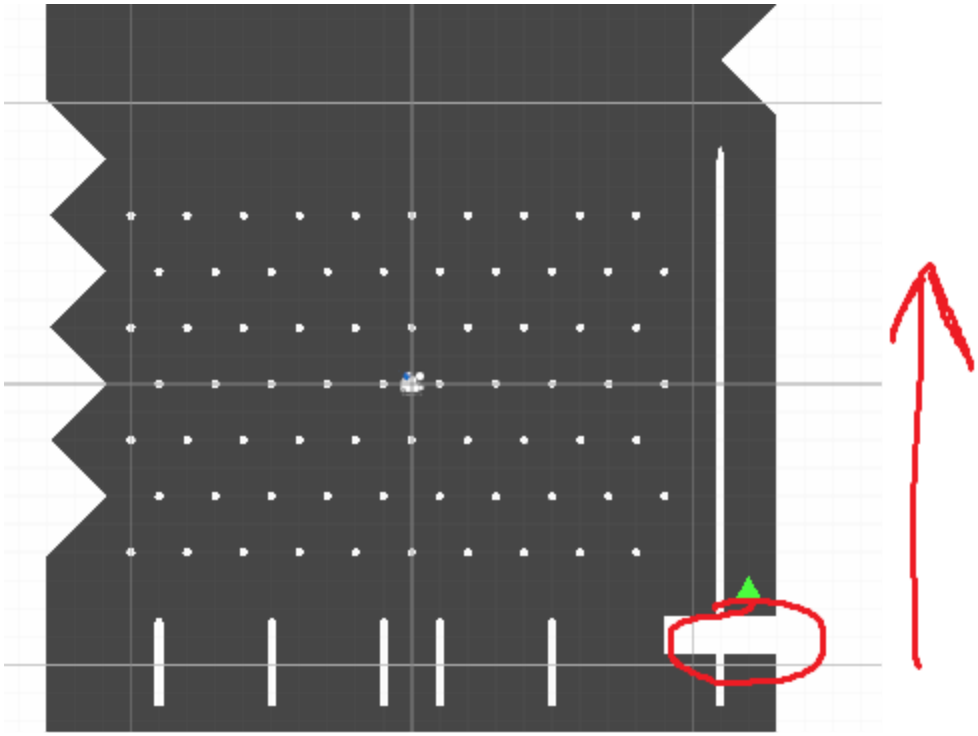
KeyCode.Space = 32
Space key.

CS1503: Argument 1: cannot convert from 'UnityEngine.KeyCode' to 'string'

Anaha to the rescue

It's GetKeyUp for keyboard. GetButtonUp is for controller.

GetButtonUp is for getting a button as a string.



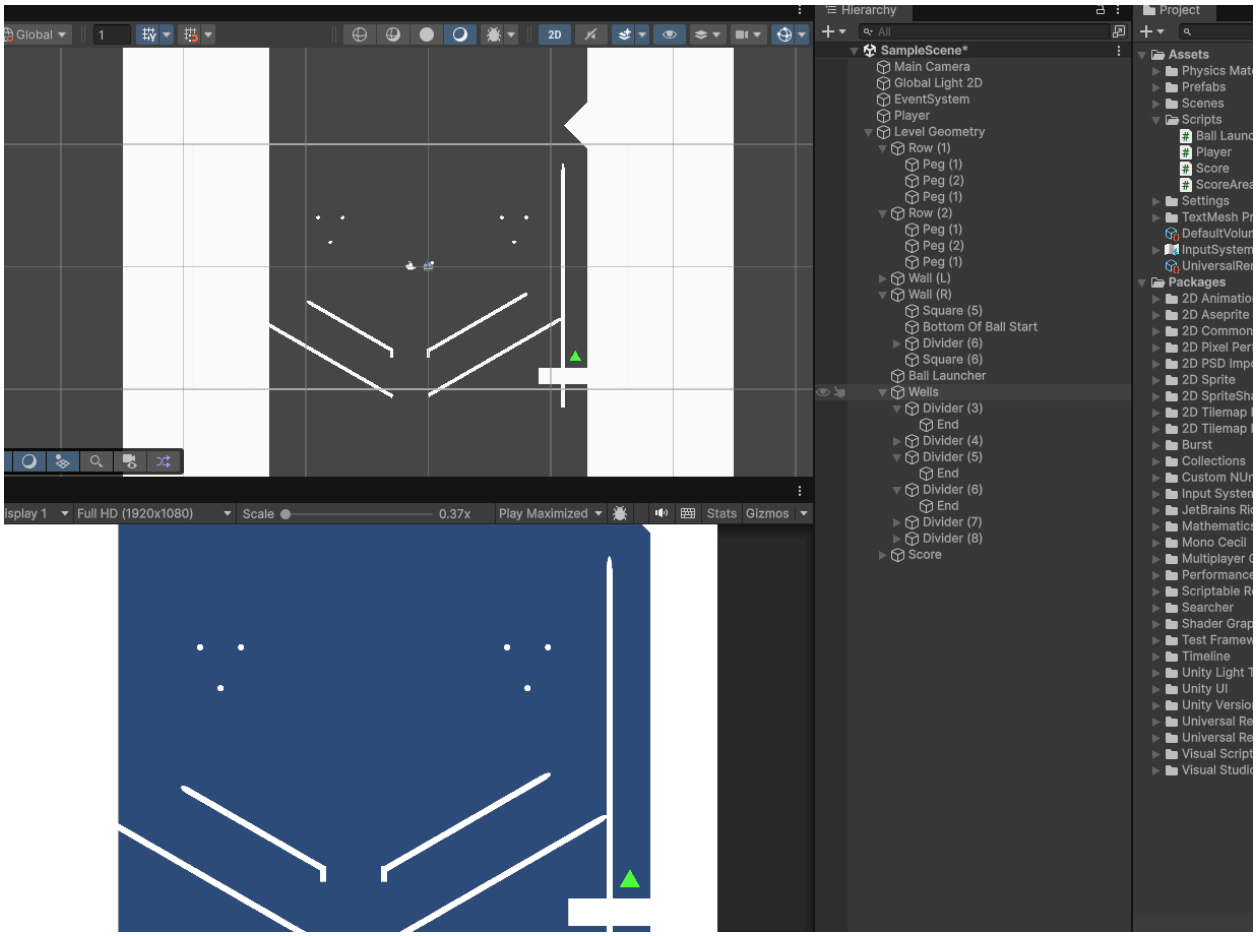
Currently stuck here, trying to figure out how to make the launcher go up. I have the code to make it return to its original position while the button isn't being held, but now I need to tie in whatever the unity framework code is for upward movement to the square.
Below is me explaining to Anaha what I'm trying to do, and asking for suggestions.

Its a ball launcher
So it needs to go up a set amount before returning to its start position
I'm 90% sure the return to origin code works
but the issue is I can't figure out how to make it just go up while holding space bar
I doubt its difficult, but Im not sure what in the framework
wait
gonna take a look

no one 5:55 PM
dose it launch or fly up?

Lek 5:56 PM
The square is meant to move upwards to launch the ball
a - la pinball
Both have a rigidbody, surely no physics bugs occur because of this
Where was the thing to open up the basic WASD control scheme in unity
the one that lets you conver strings to commands
like "horizontal" -> A / D
or "vertical" -> W / S

I still don’t have a solution. What he was saying wasn’t making a ton of sense to me. Had to go eat and so I decided to spend the rest of my time focused on making sure the initial playspace was set up.

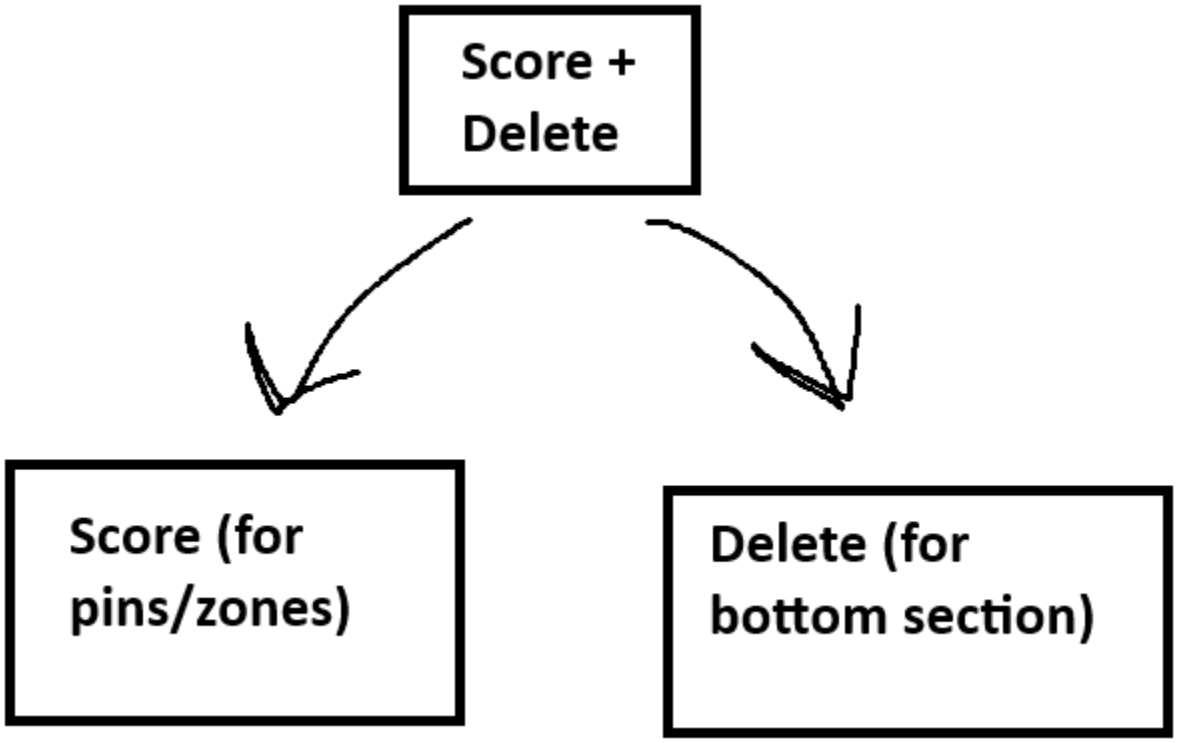


I still need to actually add the bumpers, make the pins bounce the ball, but so far it's at least *looking* like actual pinball. Here are my current theoretical solutions to those implementation challenges.

Bounce script

Essentially, the plan is to make the interior circle/existing peg have some kind of bounce modifier to repel any rigid.body that collides with it. From there, a slightly bigger circle that uses the score part of the existing score script. That neatly segways into…

Converting the score script into two divergent scripts. This would essentially allow me to save the time of figuring out how to script score mechanics and deletion mechanics, because it already exists, it just can't be in the same script for my purposes.



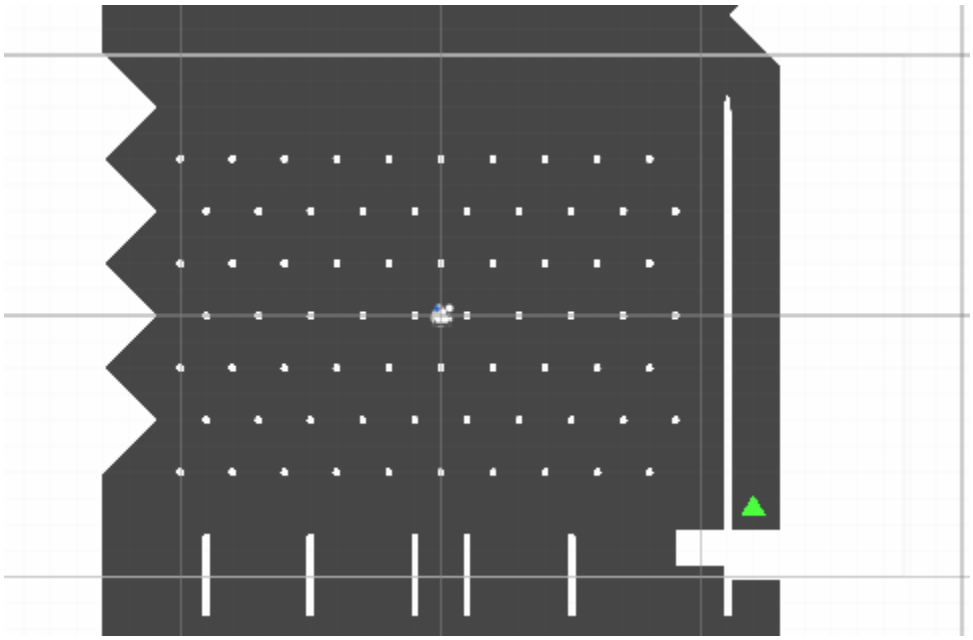
Back to implementing the ball launcher movement code, not sure what to do but currently trying this.

```
}

// Update is called once per frame
Unity Message | 0 references
void Update()
{

    // This should allow the ball to spawn and launch when spacebar is pressed
    if (Input.GetKey(KeyCode.Space))
    {
        moveY = position.y * Time.deltaTime * LaunchSpeed;
        transform.Translate(endPosition);

    }
    else
    {
```

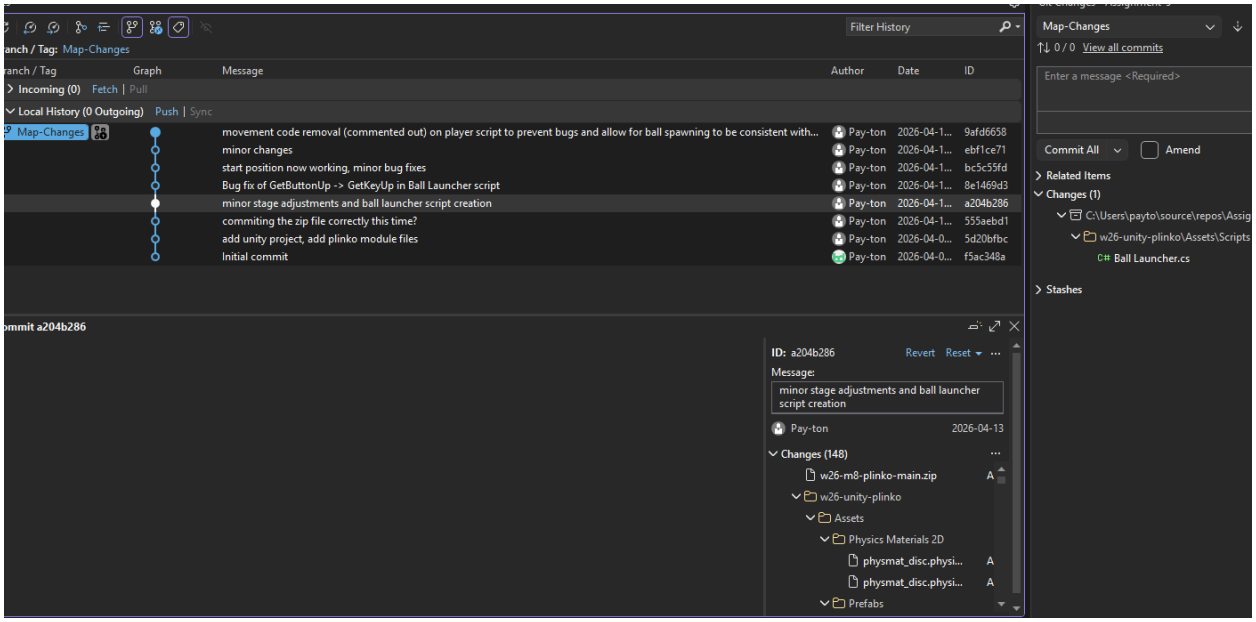
Update: As expected, it doesn't work. Worse, for some reason the translate code just teleports it instead of moving it upwards.

Honestly, Im getting close to just saying fuck it and putting the ball back to just being dropped, rather than trying to move this.

Problem is, I have no idea how the fuck to do bumpers because I was relying on the code being similar between the two.



Holy fuck I am mad right now.
So when I was last working on it, you might see a difference in that I had made stage edits, and committed them. I think because my computer for ZERO FUCKING REASON decided to randomly force windows updates, it closed my unity project on me yet apparently just didn't fucking save the changes?????? Despite me commiting them on github????



I've made an additional branch, hoping I can reset to the older version on the side branch and have things work.

Didn't work.

So now its Wednesday 11:28pm, I have no fucking clue what I am going to do at this point.

Ok so I have… news? The ball launcher works (thank you Anaha for the transform.lerp), bad news is that while it does lift the ball into the area, the area as mentioned WASN'T SAVED BECAUSE MICROSOFT IS FULL OF DICKHEADS WHO FORCE UPDATE YOUR MACHINE WITH UNSAVED WORK APPARENTLY.

Other bad news, it isn't applying a repulsion force, so the ball is actually just dropping immediately the second you stop holding the spacebar, making it just… not at all look realistic. It drops like a damn rock which doesn't feel correct. I tried messing around with the rigid body but that wasn't working either.

I want to apply some kind of repulsion force, but can't seem to work it in.

Ok, yes Anaha has assisted me again with this new bit of code that allows the repulsion to work. Apologies as I've had a lot of breakthroughs all at once and haven't been the most diligent at logging them.

```
if (Disc != null && Input.GetKey(KeyCode.Space) && InLaunchPosition.GetComponent<Collider2D>().IsTouching(Disc.GetComponent<Collider2D>()))

{

    transform.position = Vector3.Lerp(transform.position, endPosition, LaunchSpeed * Time.deltaTime);
    Disc.GetComponent<Rigidbody2D>().AddForce(new Vector2(0, 500));
}
```

Specifically, the add force to the rigid body, and the collision detection within the disc within the if statement is stuff he helped me with.

The good news is, the launcher is now working exactly as I intended! Peg collisions and score additions are the new bit I need to work in. As mentioned, the score portion should be easy.

TL ; DR - I did not actively log as I was working through the final feature implementations, so this is my recollection of today.

Anaha helped me implement two of the necessary features for the code to work as I intended, namely he helped me with greatly simplifying the code required to turn the bumpers instead of doing manual rotation inputs with grid coordinates, as well as helped me figure out collision code and part of the force.Impulse code in the launch area script which I re-used for the pegs and scoring.

I made a lot of tweaks to the level design and fine tuned a lot of the implementations as ultimately things like gravity and the bumpers felt really off to me, so I spent about an hour fiddling about with minor changes as well as completing this write up.

The video I recorded of myself playing it, as well as the read me were created before I actually finished touching up the project. I fixed things like the bumper by changing the -90 to a -45 degree based on the quaternion.

```
ScoreArea.cs    Bumpers.cs    Ball Launcher.cs    Player.cs    Git Repos
                                                  BumperLeft
nity Script (1 asset reference) | 0 references
lic class BumperLeft : MonoBehaviour


    public Quaternion startRotation;
    public Quaternion endRotation;
    public GameObject Disc;
    public GameObject BumperPosition;
    public bool flipped;


    // Start is called once before the first execution of Update after the
    Unity Message | 0 references
    void Start()
    {
        startRotation = transform.rotation;
        if (flipped)
        {
            endRotation.eulerAngles = new Vector3(0, 0, -45);
        }
        else
        {
            endRotation.eulerAngles = new Vector3(0, 0, 90);
        }
        endRotation *= startRotation;
    }
```

Ultimately, I am quite satisfied with the final touches, but obviously more can be done. Sound effects playing for the bumpers/pins?
Lives?
Alternative levels? Mini zones?
Multi ball modes?

The thing I'm most content with is the refinement, as while I am not the greatest at implementing the features I dream up, I think that I am good at solving how to fine tune them to be as I envision.